Experiment 6
EEE 3307C
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Experiment 6

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Abstract

In this experiment we learned how to design our own amplifiers, simulate them, and correctly measure all the needed parameters.

Equipment

1. Function Generator
2. Oscillioscope
3. DMM
4. Breadboard
5. Transistors
6. Resistors
7. Capacitors

Pre-Lab/Mathematical Work

Experiment 6 Final:

$V_T = 26mV$, $V_{on} = 0.7$, $\beta = 150$

Stage 1 Dc:

$$V_{B_1} = V_{cc} \cdot \frac{R_2}{R_1 + R_2} = 9 \cdot \frac{200K}{600K} = 3V$$

$$V_{E_1} = V_{B_1} - 0.7 = 3 - 0.7 = 2.3V$$

$$I_{C_1} \approx I_{E_1} = \frac{V_{E_1}}{56 + 4.7K} = \frac{2.3}{56 + 4.7K} = 0.4835996636mA$$

$$V_{C_1} = V_{cc} - I_{C_1} R_C = 9 - (0.484mA \cdot 10k\Omega) = 4.164003364V$$

Stage 2 Dc:

$$V_{B_2} = V_{cc} \cdot \frac{R_4}{R_3 \cdot R_4} = 9 \cdot \frac{47K}{47K + 47K} = 4.5V$$

$$V_{E_2} = V_{B_2} - 0.7 = 4.5 - 0.7 = 3.8V$$

$$I_{C_2} \approx I_{E_2} = \frac{V_{E_2}}{R_{e_2}} = \frac{3.8V}{2.2k\Omega} = 1.7272727 mA$$

Internal Resistance:

Stage 1: $R_{T_1} = \frac{26mV}{I_{C_1}} = \frac{26mV}{0.484mA} = 53.76347826\,\Omega$

Stage 2: $R_{T_2} = \frac{26mV}{I_{C_2}} = \frac{26mV}{1.727mA} = 15.05263182\,\Omega$

Input Resistance:

$$R_i = R_1 \parallel R_2 \parallel R_{in}$$

$$R_{in(Base1)} = 150 \cdot (53.76 + 60) = 17.066\,k\Omega$$

$$R_i = \frac{1}{\frac{1}{400k} + \frac{1}{200k} + \frac{1}{17.066k}} = 15.13\,k\Omega$$

Therefore, $R_i > 15\,k\Omega$

AC Gain:

$$R_{E_2} = R_{e_2} \parallel R_L = 2.2k\Omega \parallel 1k\Omega = 687.5\,\Omega$$

$$R_{in(Base2)} = 150 \cdot (15.05 + 687.5) = 105.383\,k\Omega$$

$$R_{in_2} = \frac{1}{\frac{1}{47k} + \frac{1}{47k} + \frac{1}{105.383k}} = 19.22\,k\Omega$$

$$A_{v_1} = \frac{R_C \parallel R_{in_2}}{\text{internal} + R_5} = \frac{\frac{10k \cdot 19.22k}{10k + 19.22k}}{53.76 + 56} = 59.92$$

$$A_{v_2} = \frac{687.5}{15.05 + 687.5} = 0.9785743713$$

$$A_v = A_{v_1} \cdot A_{v_2} = 58.63693583\ V$$

Output Resistance:

$$R_{Q_2} = R_C \parallel R_3 \parallel R_4 = \frac{1}{\frac{1}{10k} + \frac{1}{47k} + \frac{1}{47k}}$$

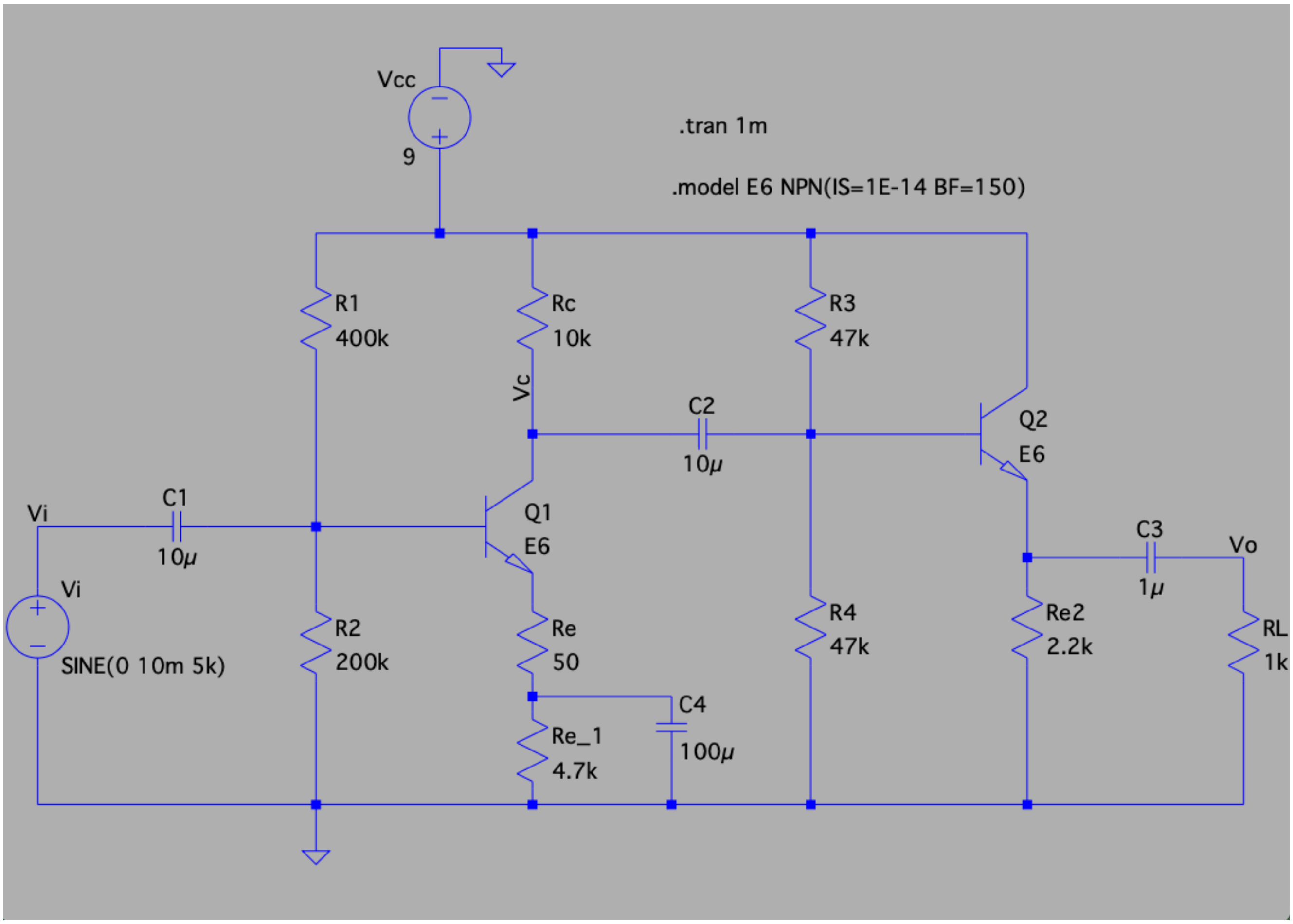
$$= 7.015k\Omega$$

$$R_{E(transistor)} = 15.05 + \frac{7.015k\Omega}{150} = 61.81880097\,\Omega$$

$$R_o = \frac{61.8 \cdot 2.2k}{61.8 + 2.2k} = 60.12920314\,\Omega$$

Circuit Designs

Design used for measurements/simulations (Abram's Design):



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Design Number 2 (Egor's Design):

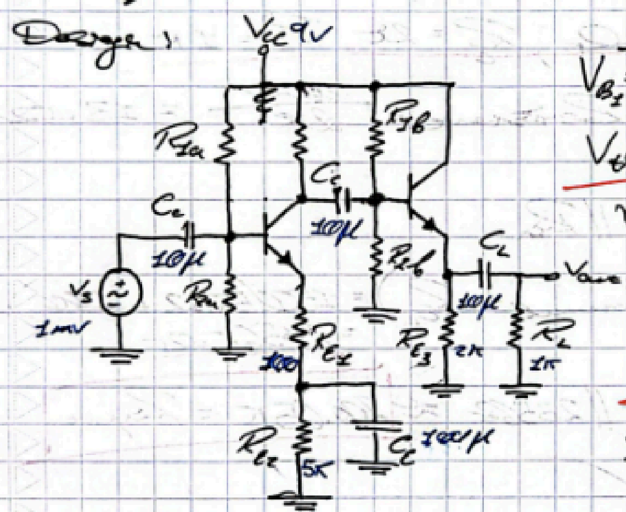
Pre-lab

Experiment 6: Design Project

Problem Statement: Design a small-signal voltage amplifier that meets the following goals

1) Voltage gain of 500 or more
2) Minimum input resistance of 15kΩ
3) Output resistance of less than 100Ω
4) Maximum undistorted output of 2v peak-to-peak
5) Power supply voltage of 9v and lower current drawn from supply voltage
6) A two-stage small-signal amplifier is more cost effective than a three-stage amplifier design

Design 1



CE

$V_{B_1} = V_{th} = V_{cc} \cdot \frac{R_2}{R_1+R_2} = 9 \cdot \frac{250K}{500K+250K}$

$V_{th} = 3V$, $V_{E_1} = 3 - 0.7 = 2.3V$

$I_{E_1} = \frac{V_{E_1}}{R_{E_1}+R_{E_2}}$

$I_{E_1} = \frac{2.3V}{5.1K} \approx 0.451mA$

$I_{C_1} \approx 0.451$

$R_{th} = \frac{R_1R_2}{R_1+R_2} = 166.67K$

$V_{th} = R_{th}I_{BQ} + I_{EQ}R_{E_1+E_2}$

$3 = (166.67K)(I_{BQ}) + (0.451mA)(5.1K)$

$0.7 \approx 166.67K(I_{BQ}) \rightarrow I_{BQ} = 4.2\mu A$

$I_{CQ_1} = 0.4468 mA$

$R_c = 16K$

$R_{1a} = 500K$

$R_{2a} = 250K$

$V_c = V_{cc} - I_cR_c$

$= 9 - (0.4468)(16K)$

$V_c = 1.8512V$

CE internal Resistance $= \frac{V_T}{I_{CQ_1}}$

$= \frac{26mV}{0.4552mA} \approx 57.12\Omega$

CE

$V_{B_2} = V_{Th} = V_{cc} \cdot \frac{R_{2b}}{R_{1b}+R_{2b}} = 9 \cdot \frac{50k}{50k+50k} = 4.5V$

$V_{E_2} = 4.5 - 0.7 = 3.8V$

$I_{EQ_2} = \frac{V_{E_2}}{R_{E_2}} = \frac{3.8V}{2K} = 1.9mA$

$I_{CQ} = \frac{150}{151}(1.9mA) \approx 1.89mA$

$R_1 = R_2 = 50k\Omega,\ R_E = 2k,\ R_L = 1K$

CE internal Resistance $= \frac{26mV}{2K} = 13\Omega$

$R_i = R_{1a} \,||\, R_{2a} \,||\, R_{in}$

$R_{in} = \beta(R_{internal} + R_{E_1}) = 150(57.12 + 1000)$
$= 23568\Omega$

$R_i = 5000k \,||\, 2500k \,||\, 23568 \approx \boxed{20.65K = R_{input}}$

$20.65k > 15k$ ✓

$R_e = R_{E2} \,||\, R_L = 2k \,||\, 1k \approx 667\Omega$

$R_{in}(CE) = 150 \cdot (13 + 667) = 102K$

$R_{in_2} = 50k \,||\, 50k \,||\, 102K \approx 20.08K$

$A_{V_1} = \frac{R_c \,||\, R_{in_2}}{R_{internal_1} + R_{E_1}} = \frac{16k \,||\, 20.08}{57.12 + 100} = 56.67$

$A_{V_2} = \frac{667}{13 + 667} = 0.98$

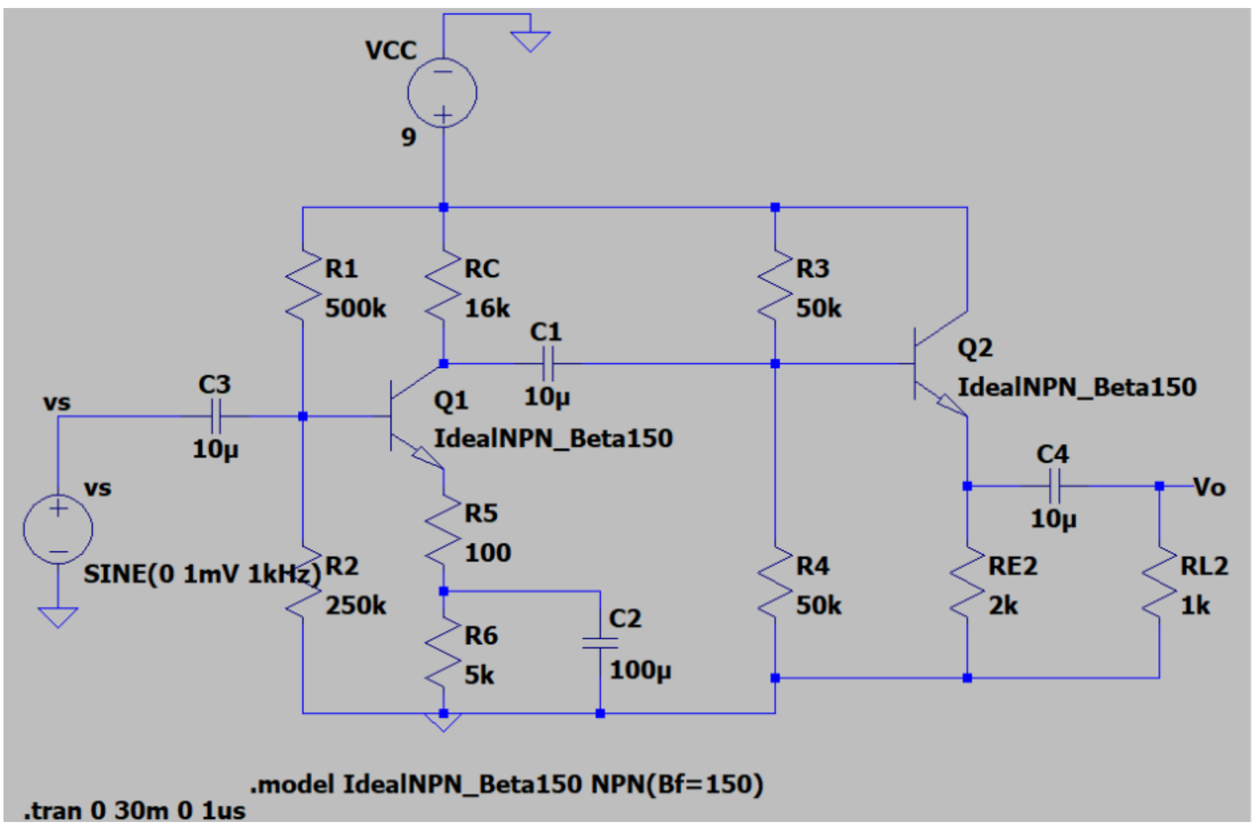
$\boxed{A_V = 56.67 \cdot 0.98 \approx 55.54}$

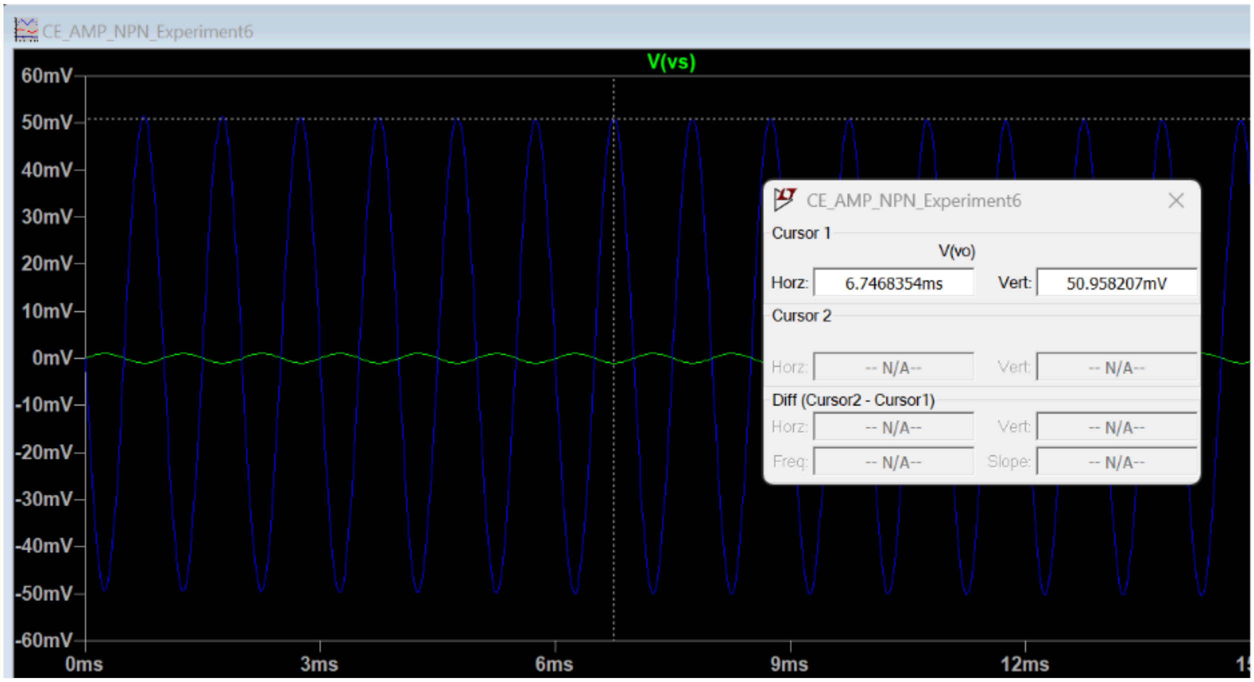
$$R_C \,||\, R_{1b} \,||\, R_{2b} = 16k \,||\, 50k \,||\, 50k \approx 9.76\,k\Omega$$

$$13 + \frac{9.76k}{150} = 78.067\,\Omega$$

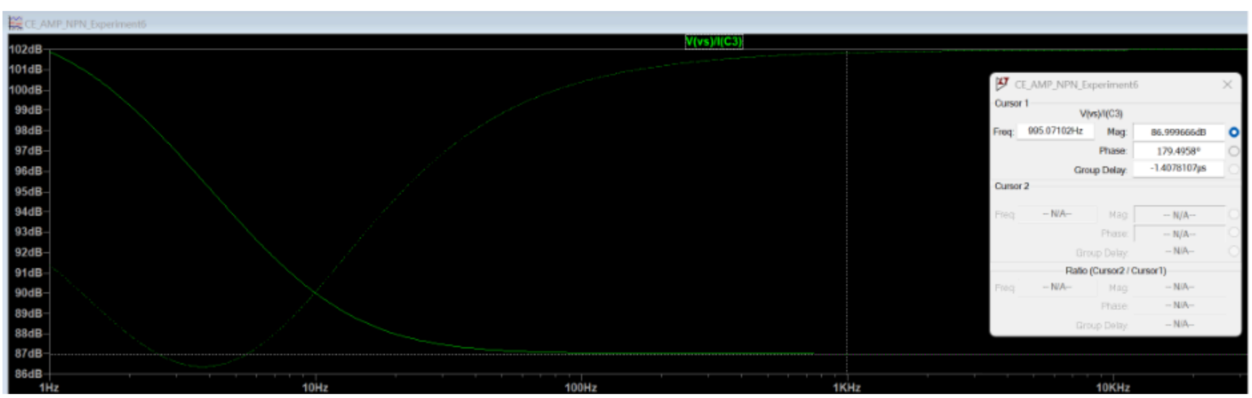
$$\boxed{R_o = \frac{78.067\,\Omega \cdot 2k}{78.067 + 2k} = 75.13\,\Omega < 100\,\Omega}$$

Simulation schematic

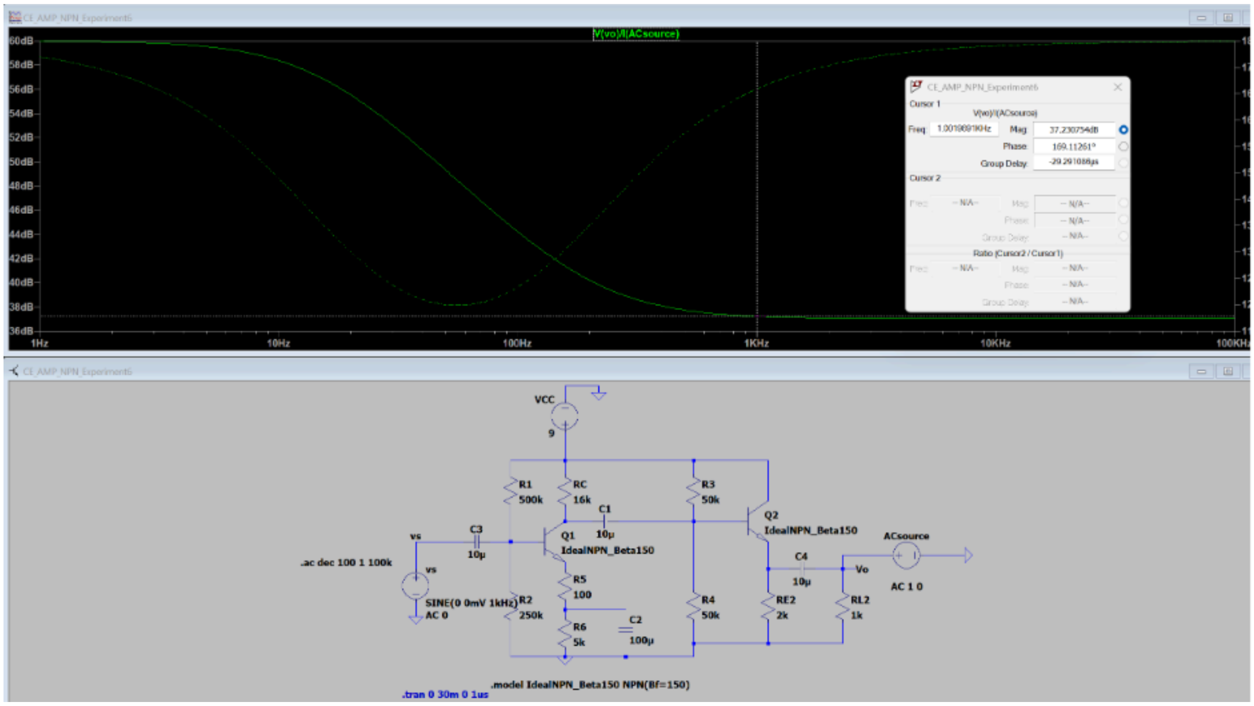




Input 1mV(green waveform) output ~51mV(blue waveform). Av ~= 51 > 50



Rin = 10^(87/20) = 22387 ohms > 15k ohms, input resistance spec met



Rout = 10^(37.2/20) = 72.44 ohms < 100 ohms, output resistance spec met

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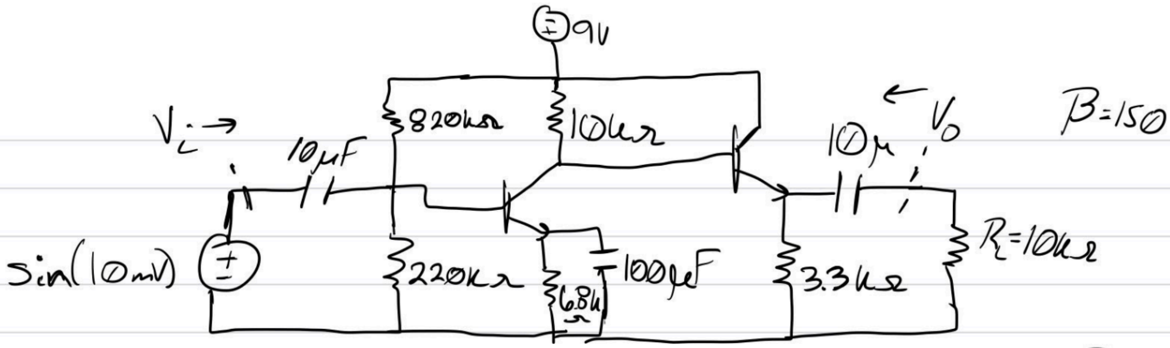
Design Number 3 (Yousef's):



$R_i = R_1 \parallel R_2 \parallel R_{in} = 820k \parallel 220k \parallel R_\pi \approx 19.6k\Omega$ ✓

$\therefore > 15k\Omega$

$R_\pi = \dfrac{150 \cdot 26mV}{\left(\dfrac{1.2V}{6.2k\Omega}\right)} = 22.2k\Omega$

1.27 1.26V

gain unloaded

$A_V = \dfrac{1.27V}{20mV} = 63.5$ ✓ $\therefore > 50$

$R_O = \dfrac{R_L (V_o - V_L)}{V_L}$ $V_L \rightarrow 1.26$

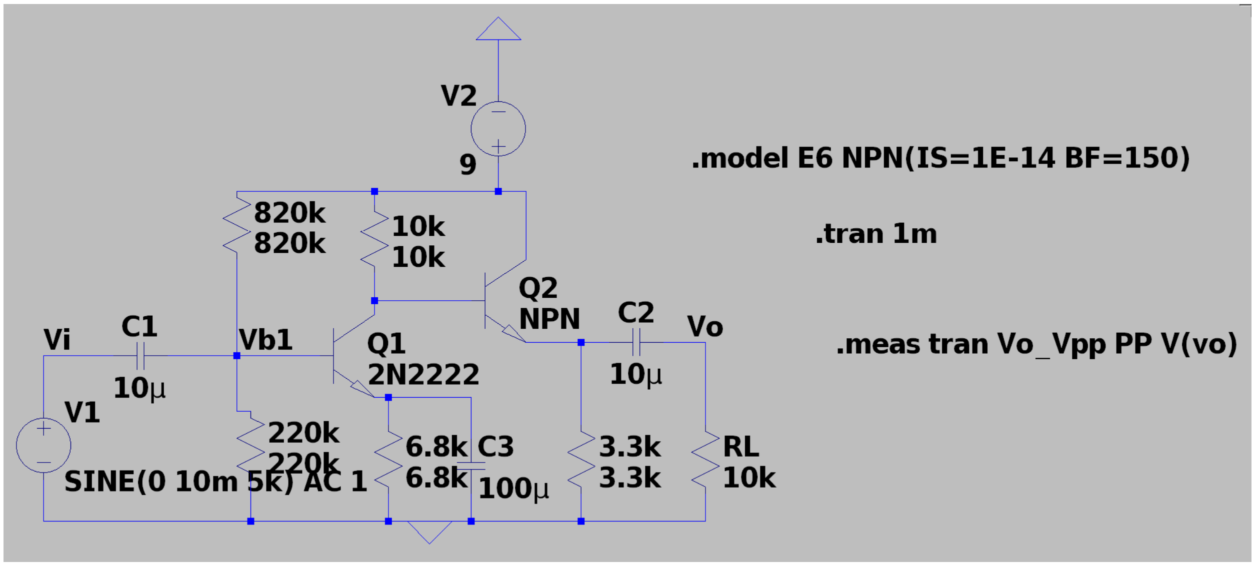
$A_{V_L} = \dfrac{1.26V}{20mV} = 63$ $\therefore > 50$

gain loaded

$R_o = \dfrac{10k \cdot (1.27 - 1.26)}{1.26} = 79.365\Omega$ ✓

$\therefore < 80\Omega$

Design Number 3 LtSpice (Yousef's):



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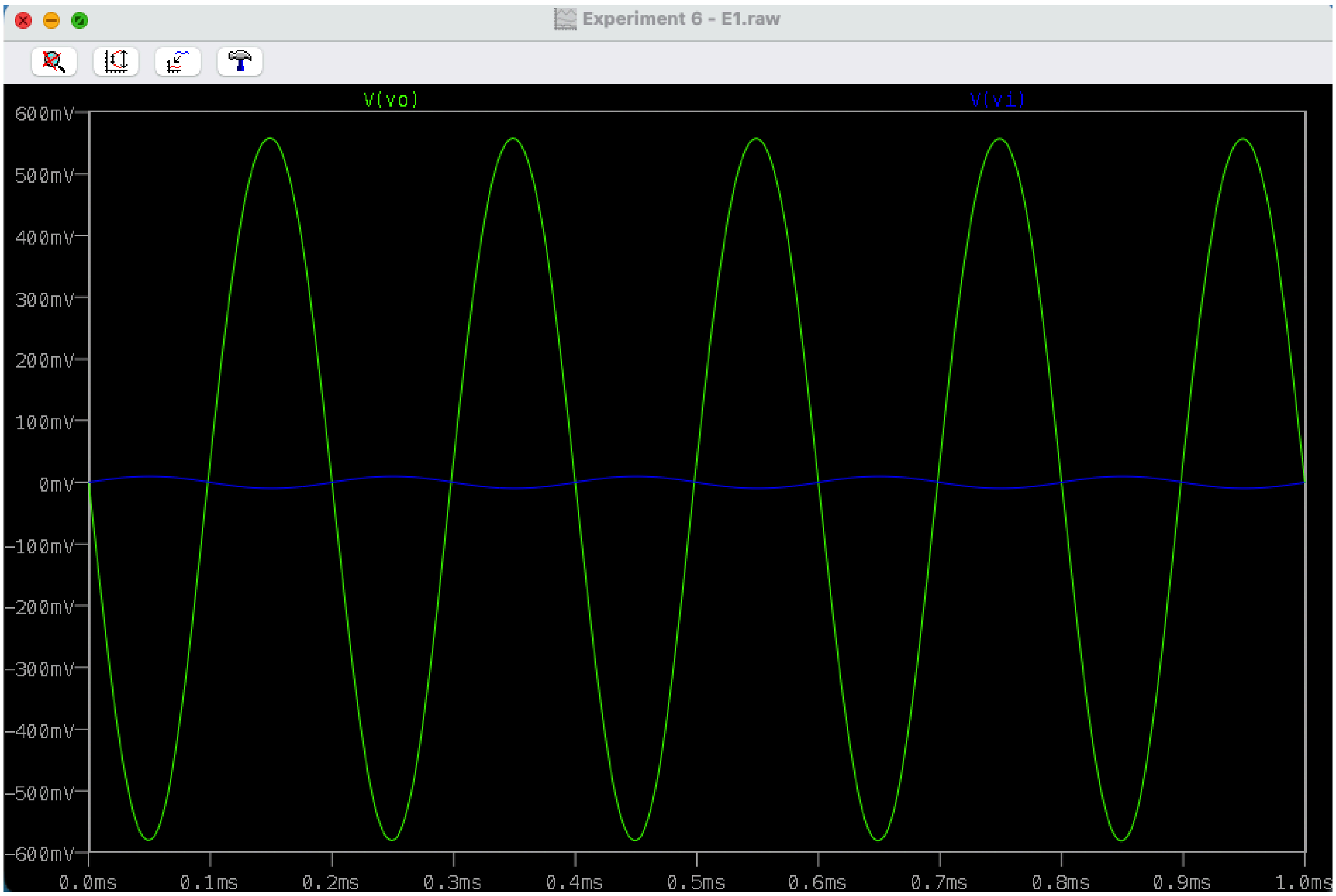
Simulation (For Design Used in Experiment)

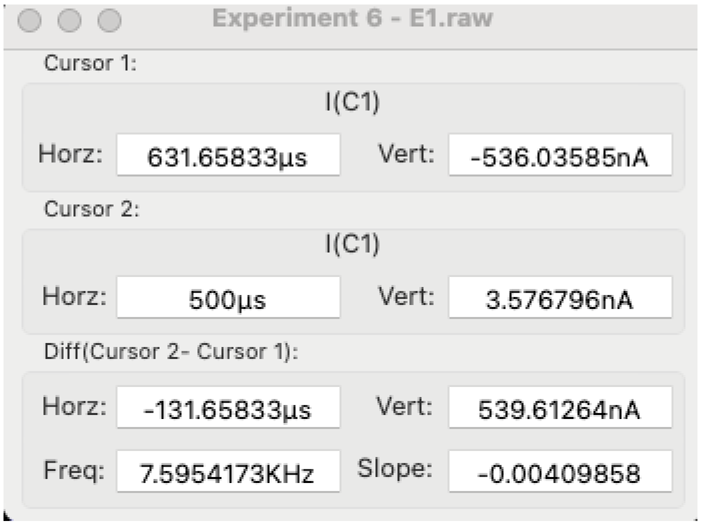
Part 1

Voltage Gain of around 55mV:



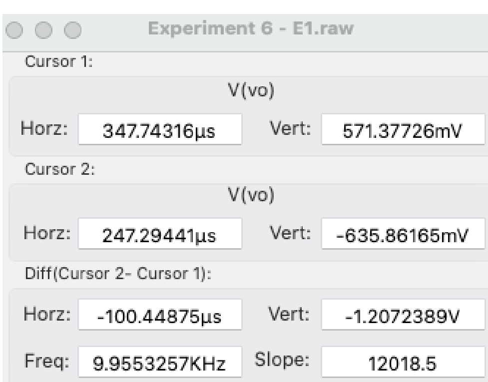
Part 2

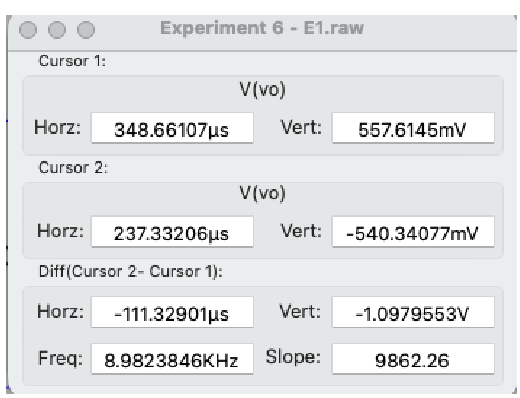
Input Resistance given these values obtained from LTspice. We can calculate R_I = 10mV / 536.04nA = 18.655 kΩ which is greater than 15 kΩ



Part 3

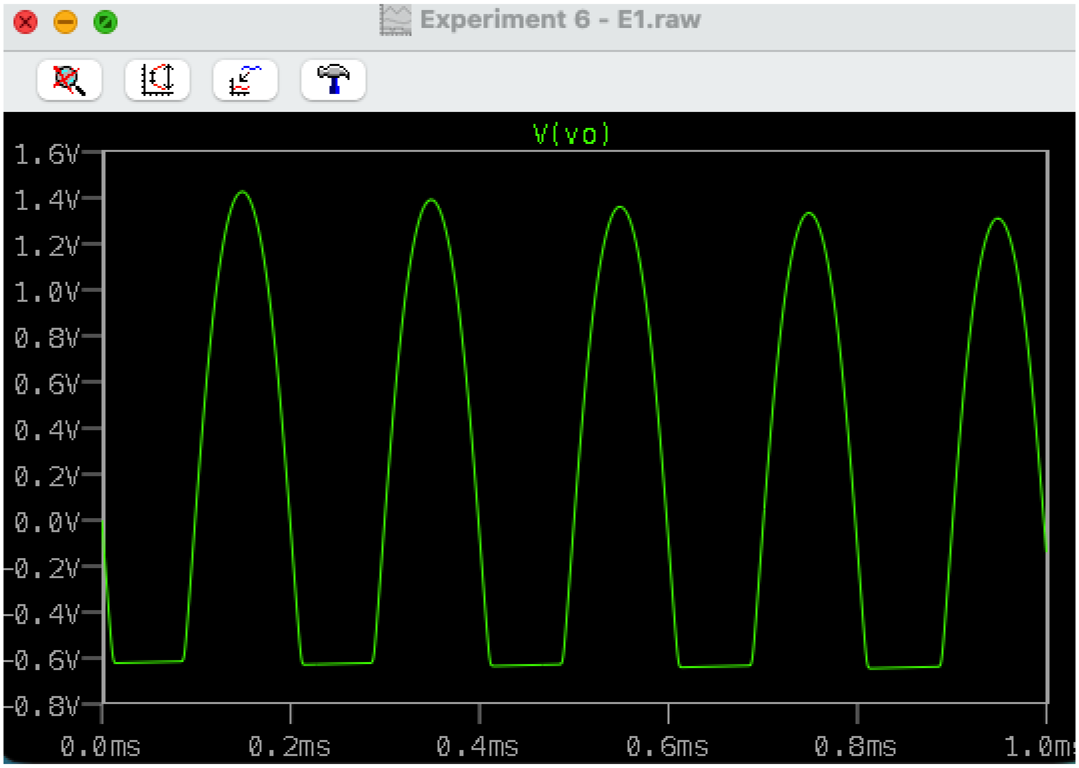
To achieve the required output resistance we first measure V_{out} with R_L connected and with it disconnected. Finally, we used this formula to calculate it, R_{OUT} = 1k * ($V_{Disconnected}$/$V_{Connected}$ - 1) = 1000 * ((1.21/1.09) -1) = 110 Ω





Part 4

Tested the maximum unclipped voltage of 1.4 - (-0.6) = 2V



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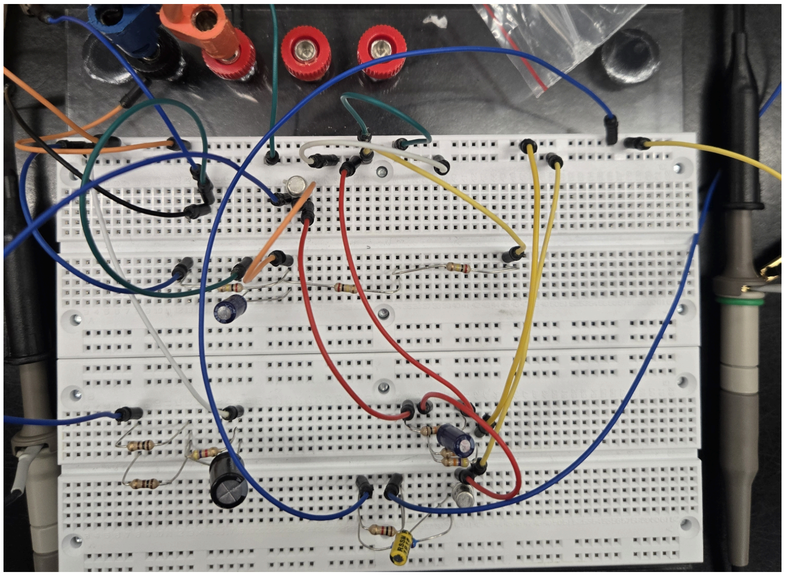
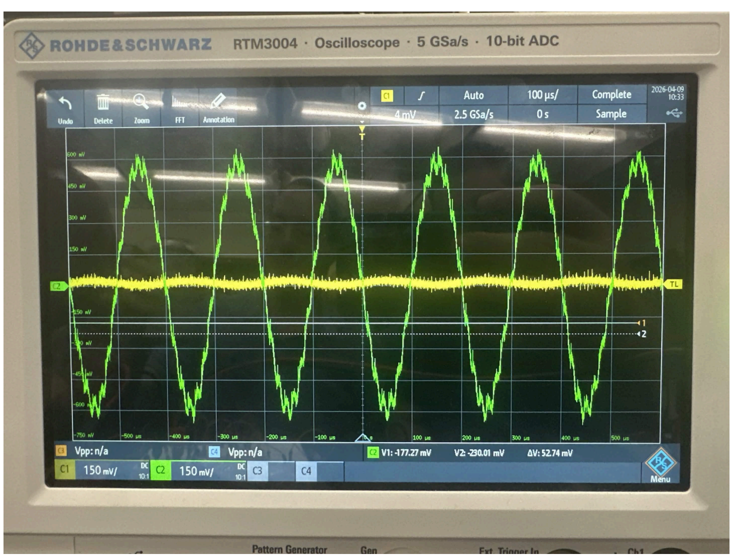
Experiment Procedure

Breadboard Circuit:



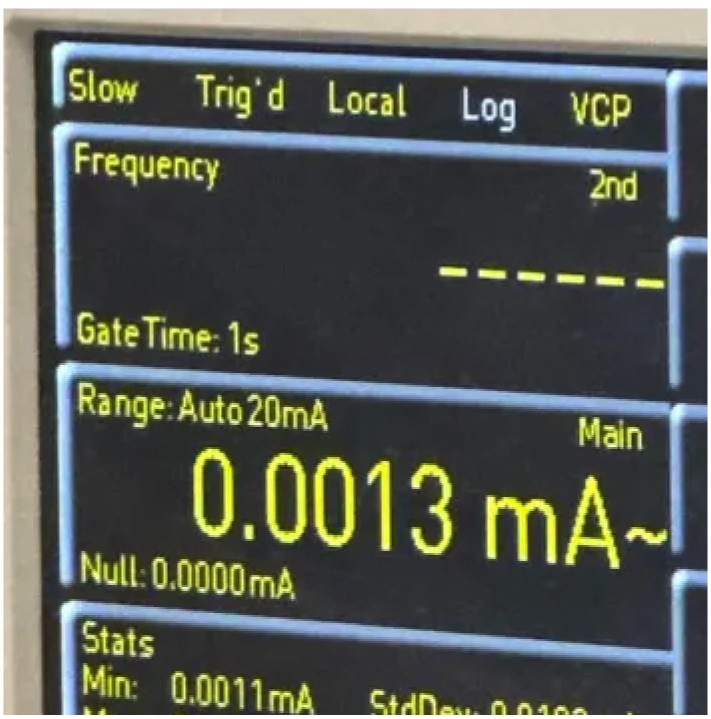
Part 1

In the first part of the lab, we used the oscilloscope in order to measure the voltage gain which was measured at around 52.76 mV.



Part 2

In the second part of the lab, we used the V_{IN} and I_S values to measure input resistance. We used 60 mV for V_{IN}, which results in about 26k ohms.



Part 3

In the third part of the lab, we measured the output resistance using

R_{OUT} = 1k * ($V_{Disconnected}/V_{Connected}$ - 1) = 1000 * ((1.21/1.09) -1) = 110 Ω

Part 4

In the fourth part of the lab, we utilized the oscilloscope to measure the clipped and unclipped voltage at V_{IN} = 20 mV and V_{IN} = 30 mV.

This picture is for V_{IN} = 20 mV for unclipped voltage.

This picture is for V_{IN} = 30 mV for clipped voltage.

Conclusion

In conclusion, in this design lab, we learned how to use our analysis skills that we have gained during the semester to finally design a two-stage AC Transistor amplifier to achieve the design requirements given to us.